

Alexander Christhian

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Education

Universitas Indonesia, BS in Computer Engineering July 2023 – Present

- GPA: 3.75/4.0
- Focusing on hardware-software integration, embedded systems, digital design, and AI
- Relevant Coursework: Computer Architecture, Digital Systems Design, Data Structures and Algorithms

Experience

Digital Laboratory Assistant, Department of Electrical Engineering FTUI 2025 – Present

- Assigned as PIC of Digital System Design laboratory sessions
- Helping in various laboratory practicum activities
- Supporting students in understanding digital logic design and VHDL programming

Programming Division Staff, AUAV VTOL Team Feb 2025 – Dec 2025

- Working on autonomous VTOL aircraft programming
- Assigned on developing QR code-based navigation system
- Assigned on implementing YOLOv11 for object detection

Staff of Academic and Profession, Ikatan Mahasiswa Elektro (IME) Feb 2024 – Dec 2024

- Assigned as PIC of Sharing Registrasi 2025 event
- Assigned on creating Fundamentals of Digital Systems, Computer Organization, and Statistics modules
- Supporting various academic and career development programs

Projects

PIP FTUI RAG System

- AI-powered academic document assistant for FTUI students using RAG architecture
- Implemented semantic search using ChromaDB and SentenceTransformer for document retrieval
- Real-time streaming chat interface with Groq LLM (Llama 3.1 70B)
- Collaborated with DAFFAsd and vinend on development
- Technologies: Python, FastAPI, Next.js, TypeScript, ChromaDB, Tailwind CSS

The Game Library github.com/vinend/PAS-02

- Comprehensive game management system built in C with custom memory allocation and data structures
- Developed interactive CLI with advanced features for game organization
- Collaborated with vinend on implementation and testing
- Technologies: C, Data Structures, CLI

Digilab-NG github.com/AlexanderChristhian/Digilab-NG

- Full-stack educational platform prototype with real-time collaboration features using Redis
- Implemented containerized deployment with Docker for scalability and ease of deployment
- Collaborated with DAFFAsd on frontend and backend development
- Technologies: React, TypeScript, Express, PostgreSQL, Redis, Docker, Cloudinary, Node.js

Mie Babi Rodotua github.com/AlexanderChristhian

- E-commerce platform for traditional cuisine with cloud-based image storage and optimization

- Implemented serverless deployment on Vercel for optimal performance
- Technologies: React, Tailwind, Vite, JavaScript, Express, NeonDB, PostgreSQL, Cloudinary

RTOS Scheduler in VHDL

github.com/AlexanderChristhian

- Real-time operating system scheduler with hardware implementation and custom datapath
- Implemented task control block memory management for efficient task scheduling
- Technologies: VHDL, Vivado, FPGA, RTOS

8x8 Snake Game Using AVR Assembly

github.com/AlexanderChristhian

- Classic snake game on 8x8 LED matrix with low-level AVR assembly programming
- Implemented hardware simulation in Proteus for testing and validation
- Technologies: Arduino, AVR, Assembly, Proteus

Technical Skills

Languages: C, C++, C#, Python, JavaScript, TypeScript, Java, VHDL, Assembly

Frontend: React, Next.js, Tailwind CSS, HTML5, CSS3, Vite

Backend & Database: Node.js, Express.js, FastAPI, PostgreSQL, MongoDB, Redis, NeonDB, Cloudinary

DevOps & Cloud: Docker, Vercel, Ansible, Alibaba Cloud

Hardware & FPGA: VHDL, Assembly, Proteus, Vivado, Quartus Prime, ModelSim

Embedded & IoT: Arduino, ROS2, MQTT, FreeRTOS, PX4 Autopilot

AI & ML: ChromaDB, Groq, RAG, SentenceTransformer